

AI Debt Relief Platform

Project Description:

AI Debt Relief Platform is an intelligent web-based financial assistance platform designed to help individuals effectively manage and reduce their debt through AI-powered guidance. The platform leverages the **Google Gemini API** to analyze users' financial information, including income, expenses, existing loans, and repayment capacity, and generates personalized debt repayment strategies and financial recommendations.

The system features secure user authentication using **JWT**, allowing users to create accounts, manage their financial profiles, and access personalized debt management plans. It also provides AI-driven insights, repayment prioritization, budgeting suggestions, and financial education resources to help users make informed financial decisions.

Built with the **MERN stack** (React.js, Node.js, Express.js, and MongoDB Atlas), the platform offers a responsive and user-friendly interface. The frontend is deployed on **Vercel**, while the backend is hosted on **Render**, ensuring scalability, reliability, and seamless accessibility. By combining modern web technologies with Generative AI, the AI Debt Relief Platform empowers users to regain financial stability, reduce debt efficiently, and improve long-term financial well-being.

Scenarios

Scenario 1: Debt Repayment Planning

A user registers on the platform and enters their monthly income, expenses, outstanding loans, interest rates, and repayment deadlines. The AI analyzes the financial data and generates a personalized debt repayment strategy, prioritizing high-interest debts while suggesting an affordable monthly repayment plan.

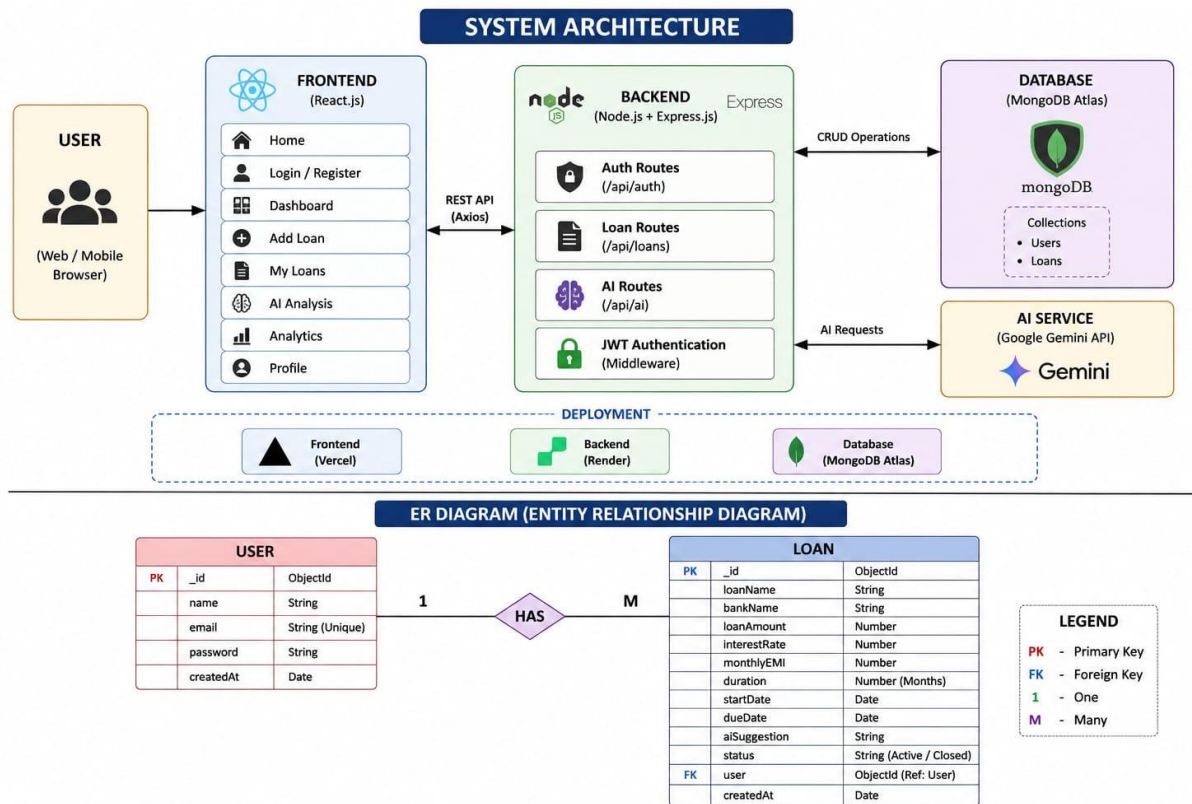
Scenario 2: Budget Optimization and Financial Guidance

A user struggling with monthly expenses uploads their financial details to the platform. The AI evaluates spending patterns, identifies unnecessary expenses, and provides a customized budget plan with practical saving tips. It also recommends ways to allocate income efficiently to accelerate debt repayment.

Scenario 3: Loan Comparison and Debt Relief Assistance

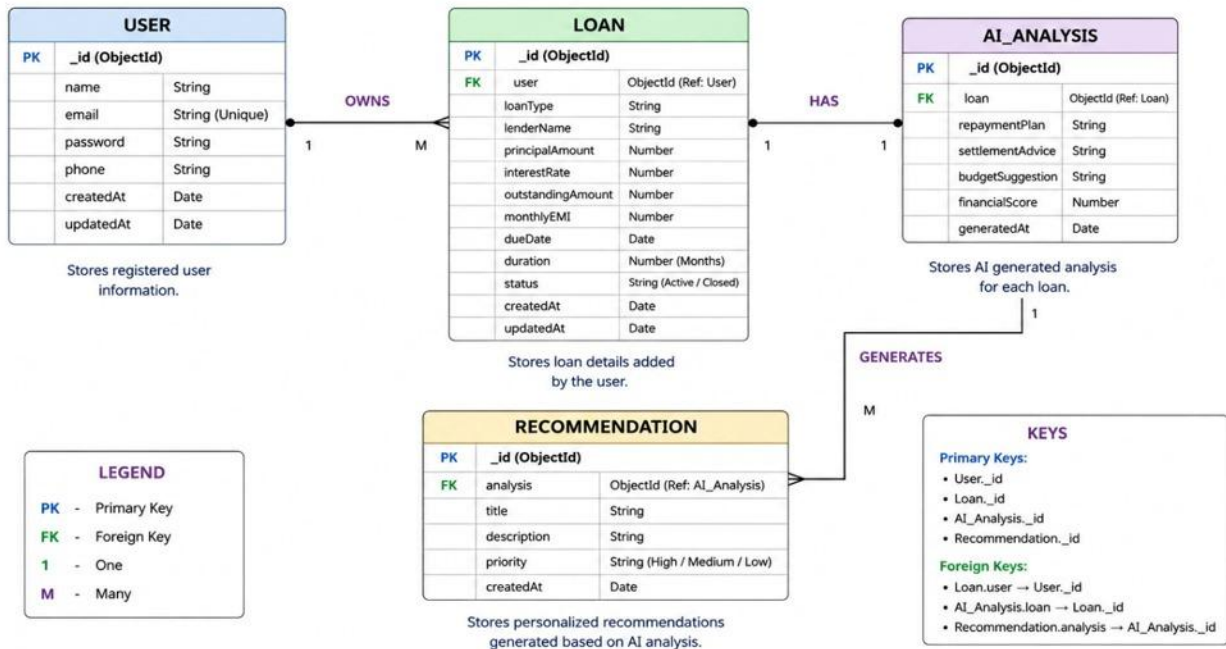
A user is looking for better repayment options for existing loans. The platform analyzes the user's financial profile and current debts, then suggests suitable debt relief strategies such as loan consolidation, refinancing opportunities, or structured repayment plans. The AI also explains the benefits and potential risks of each option, enabling the user to make informed financial decisions.

Technical Architecture:



Description: The **AI Debt Relief Platform** follows a secure **three-tier architecture** consisting of a React.js frontend, a Node.js/Express.js backend, and a MongoDB Atlas database with Google Gemini AI integration. The frontend provides a responsive interface where users can securely register, log in, and enter their financial information. The Express.js backend handles authentication using JWT, validates user inputs, manages API requests, and communicates with both the MongoDB database and the Google Gemini API.

ER DIAGRAM – AI DEBT RELIEF PLATFORM



ER DIAGRAM DESCRIPTION

The AI Debt Relief Platform database consists of four main entities: User, Loan, AI_Analysis, and Recommendation. A user can add multiple loans (1:M). Each loan is analyzed by the AI engine to generate a detailed financial analysis (1:1). Based on this analysis, one or more personalized recommendations are generated (1:M).

Pre-requisites:

HTML, CSS, and JavaScript Skills: <https://www.w3schools.com/>

JavaScript Programming Knowledge: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>

React.js Framework Knowledge: <https://react.dev/learn>

Node.js Development: <https://nodejs.org/en/docs>

Express.js Framework: <https://expressjs.com/>

MongoDB Atlas Database: <https://www.mongodb.com/docs/atlas/>

Mongoose ODM: <https://mongoosejs.com/docs/>

JWT Authentication: <https://jwt.io/introduction>

Axios HTTP Client: <https://axios-http.com/docs/intro>

Google Gemini API: <https://ai.google.dev/gemini-api/docs>

Git Version Control: <https://git-scm.com/doc>

GitHub Repository Management: <https://docs.github.com/>

Visual Studio Code Setup: <https://code.visualstudio.com/docs>

Vercel Frontend Deployment: <https://vercel.com/docs>

Render Backend Deployment: <https://render.com/docs>

Postman API Testing: <https://learning.postman.com/docs/>

Project Workflow:

Milestone 1: Project Planning and System Architecture

- **Activity 1.1:** Define the project objectives, scope, and functional requirements for the AI Debt Relief Platform.
- **Activity 1.2:** Select the technology stack, including React.js, Node.js, Express.js, MongoDB Atlas, and Google Gemini API.
- **Activity 1.3:** Design the overall system architecture, illustrating interactions between the frontend, backend, database, authentication, and AI services.
- **Activity 1.4:** Set up the development environment by installing the required software, libraries, and dependencies.

Milestone 2: Backend Development

- **Activity 2.1:** Develop RESTful APIs using Node.js and Express.js for user management and financial data processing.
- **Activity 2.2:** Implement JWT-based authentication for secure user registration and login.
- **Activity 2.3:** Configure MongoDB Atlas using Mongoose to store user profiles, financial information, debt details, and AI-generated recommendations.
- **Activity 2.4:** Integrate the Google Gemini API to analyze financial data and generate personalized debt relief recommendations.

Milestone 3: Frontend Development

- **Activity 3.1:** Design and develop a responsive user interface using React.js, HTML, CSS, and JavaScript.
- **Activity 3.2:** Create pages for user registration, login, dashboard, financial data entry, debt management, and AI recommendation display.
- **Activity 3.3:** Connect the frontend with backend APIs using Axios and implement real-time data rendering.

Milestone 4: AI Integration and Core Functionalities

- **Activity 4.1:** Process user financial information and securely send it to the backend.
- **Activity 4.2:** Generate AI-powered debt repayment strategies, budgeting suggestions, and financial insights using Google Gemini.
- **Activity 4.3:** Display personalized recommendations and repayment plans in an easy-to-understand format.
- **Activity 4.4:** Validate user inputs and implement proper error handling for reliable system performance.

Milestone 5: Testing and Deployment

- **Activity 5.1:** Perform unit testing, integration testing, and functional testing to verify system performance.
- **Activity 5.2:** Fix identified bugs and optimize application performance.
- **Activity 5.3:** Deploy the React.js frontend on **Vercel** and the Node.js/Express.js backend on **Render**.
- **Activity 5.4:** Verify database connectivity, API communication, AI integration, and overall application functionality after deployment.

Milestone 6: Conclusion

- Evaluate the overall performance of the AI Debt Relief Platform and verify that all functional requirements have been successfully implemented. The completed system provides secure user authentication, AI-powered financial analysis, personalized debt repayment strategies, budgeting recommendations, and a responsive user experience, while offering a scalable foundation for future enhancements.

Milestone 1: Project Planning and System Architecture

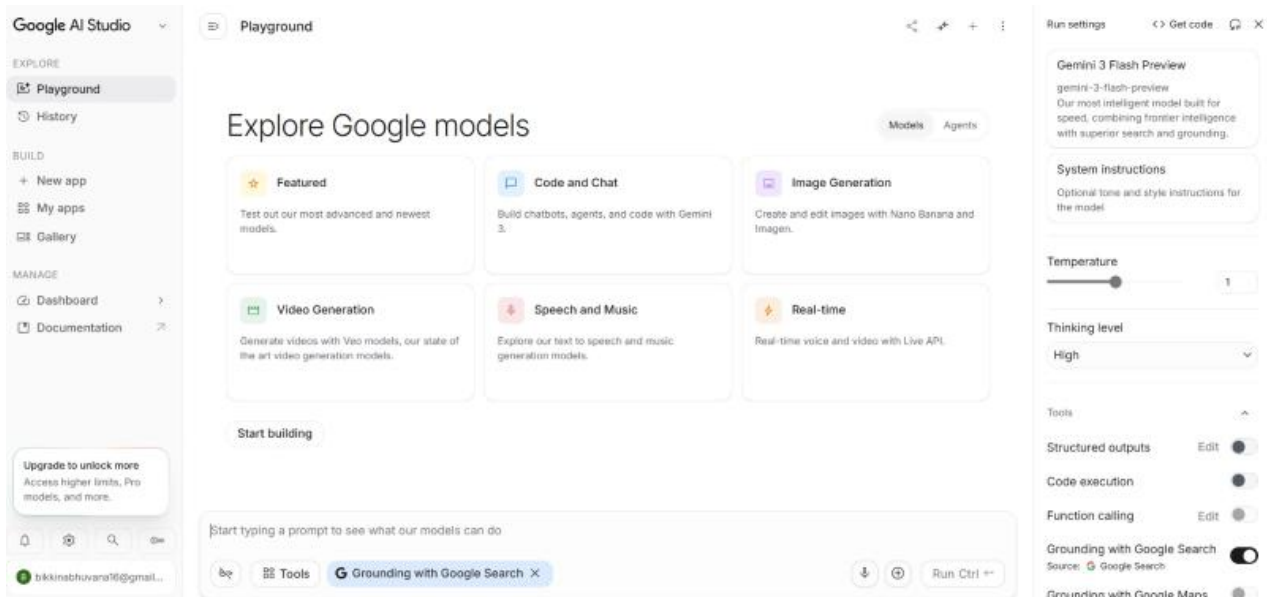
In this milestone, the focus is on planning the AI-Powered Debt Relief & Financial Recovery Platform by selecting the appropriate technologies, designing the system architecture, configuring the development environment, and integrating Google Gemini AI. The objective is to build a secure, scalable, and intelligent web application that provides personalized debt management and financial recovery recommendations.

Activity 1.1: Generate a Google Gemini API Key

Before the application can communicate with the Google Gemini Large Language Model (LLM), a valid Gemini API key is required. Follow the steps below to generate and configure the API key securely.

Step 1 – Create a Google AI Studio Account

Visit <https://aistudio.google.com/> and sign in using your Google account.



Step 2 – Navigate to API Keys

After logging in, click "Get API Key" from the left-side menu or navigate to the API Keys section.



Step 3 – Generate a New API Key

Click "Create API Key", select your Google Cloud project (or create a new one if required), and generate the API key. Copy the generated API key and store it securely.

Step 4 – Configure the API Key

Create a .env file inside the backend project directory and add the following environment variables:

PORT=5000

MONGO_URI=your_mongodb_connection_string

JWT_SECRET=your_jwt_secret

GEMINI_API_KEY=your_google_gemini_api_key

Step 5 – Load Environment Variables

Install dotenv if it is not already installed.

```
npm install dotenv
```

Import dotenv in the backend server.

```
import dotenv from "dotenv";
```

```
dotenv.config();
```

Access the API key.

```
const apiKey = process.env.GEMINI_API_KEY;
```

Important:

- Never upload your .env file to GitHub.
- Add the following line to, gitignore.env

```
server > .env
1  PORT=5000
2
3  MONGO_URI=mongodb+srv://23pa1a0529_db_user:bhuvana1685@cluster0.bbj8cfx.mongodb.net/
  debtre relief?retryWrites=true&w=majority&appName=Cluster0
4
5  JWT_SECRET=DebtReliefSecret123
6  GEMINI_API_KEY=AQ.Ab8RN6J5fyoiayV-U3dlPaMzG27LVliIkCmT_QyXn_mvNmJQ
```

Activity 1.2: Research and Select the Appropriate Technologies

Before beginning development, different technologies were evaluated based on performance, scalability, ease of development, security, and compatibility with AI integration.

The following technologies were selected for the project.

- Frontend
- React.js
- HTML5
- CSS3
- JavaScript
- Axios
- Chart.js

React.js was selected because it enables component-based development, reusable UI components, fast rendering through the Virtual DOM, and responsive user interfaces.

- Backend

- Node.js
- Express.js

Node.js provides fast and efficient server-side execution, while Express.js simplifies REST API development and routing.

- Database
- MongoDB Atlas
- Mongoose ODM

MongoDB Atlas was selected because it provides a scalable cloud-hosted NoSQL database capable of efficiently storing user information, loan details, and AI-generated financial recommendations.

- Authentication
- JSON Web Token (JWT)
- bcrypt.js

JWT enables secure authentication and authorization, while bcrypt.js encrypts user passwords before storing them in the database.

- Artificial Intelligence
- Google Gemini API

Google Gemini AI was selected because it provides powerful natural language understanding and generation capabilities. It analyzes users' financial information and generates personalized debt settlement strategies, budgeting recommendations, financial insights, and professional negotiation letters.

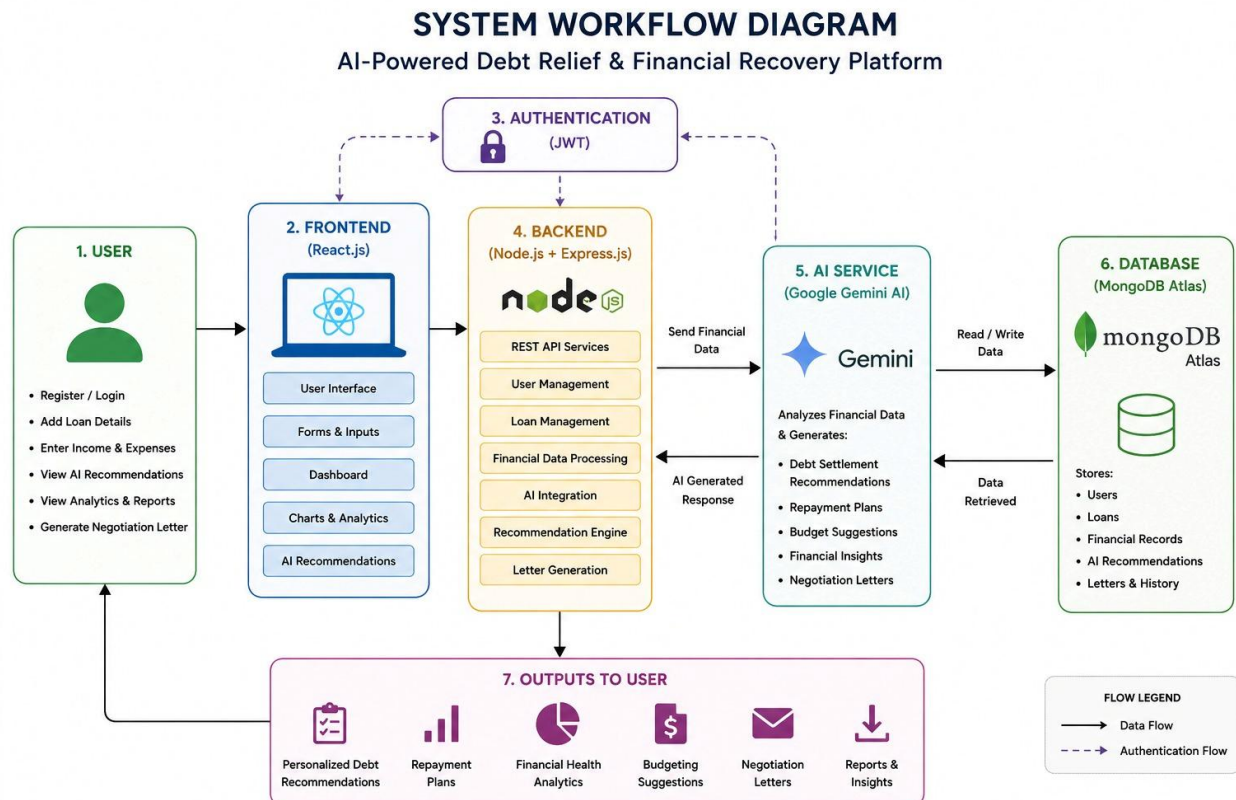
- Development Tools
- Visual Studio Code
- Git
- GitHub
- Postman
- MongoDB Compass

These tools assist in coding, version control, API testing, and database management throughout the development process.

- Reasons for Selecting the Technology Stack
- React.js enables fast and interactive user interfaces.
- Node.js and Express.js provide efficient backend processing.
- MongoDB Atlas supports scalable cloud-based data storage.
- Google Gemini AI delivers intelligent financial recommendations.
- JWT ensures secure user authentication.
- Chart.js provides interactive financial analytics and visualizations.

Activity 1.3: Define the System Architecture

The AI-Powered Debt Relief Platform follows a three-tier architecture, consisting of the Presentation Layer, Business Logic Layer, and Database Layer. The architecture enables secure communication between users, backend services, AI components, and the cloud database.



Frontend Layer

The frontend is developed using React.js and provides a responsive and user-friendly interface.

Its responsibilities include:

- User Registration
- User Login
- Dashboard
- Loan Management
- AI Analysis
- Analytics Dashboard
- Financial Reports

- Backend Layer

The backend is developed using Node.js and Express.js.

Its responsibilities include:

- User Authentication
- Loan CRUD Operations
- Financial Data Processing
- REST API Development
- AI Integration
- Recommendation Generation
- Negotiation Letter Generation
- Database Layer

MongoDB Atlas stores all application data securely.

Collections include:

- Users
- Loans
- AI Recommendations
- Financial Records
- AI Integration

The backend securely sends the user's financial information to Google Gemini AI.

The AI analyzes:

- Monthly Income
- Monthly Expenses
- Loan Amount
- EMI
- Interest Rate
- Due Dates
- Outstanding Balance

Based on the analysis, Gemini AI generates:

- Debt Settlement Recommendations

- Repayment Plans
- Budget Suggestions
- Financial Health Analysis
- Negotiation Letters

Activity 1.4: Set Up the Development Environment

Before implementing the application, configure the development environment.

Step 1 – Install Node.js

Download and install the latest LTS version of Node.js from the official website.

Verify installation.

```
C:\Users\Bhuvanabikkina\OneDrive\Desktop\AI-Debt-Relief-Platform>node -v
v22.21.0

C:\Users\Bhuvanabikkina\OneDrive\Desktop\AI-Debt-Relief-Platform>npm -v
10.9.4
```

Step 2 – Install Visual Studio Code

Download and install Visual Studio Code.

Install recommended extensions such as:

- ES7 React Snippets
- Prettier
- ESLint
- MongoDB Extension

Step 3 – Clone the Repository: Clone the project repository.

```
</> Bash

git clone https://github.com/your-username/AI-Debt-Relief-Platform.git
```

Navigate to the project directory

```
</> Bash

cd AI-Debt-Relief-Platform
```

Step 4 - Install Backend Dependencies: Navigate to the backend folder.

```
C:\Users\Bhuvanabikkina\OneDrive\Desktop\AI-Debt-Relief-Platform>npm install

up to date, audited 371 packages in 7s

85 packages are looking for funding
  run `npm fund` for details

31 vulnerabilities (1 low, 17 moderate, 9 high, 4 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
```

Step 5 - Install Frontend Dependencies

Navigate to the frontend folder.

```
</> Bash

cd frontend

npm install
```

Step 6 - Configure Environment Variables

Create the .env file.

```
server > .env
1  PORT=5000
2
3  MONGO_URI=mongodb+srv://23pa1a0529_db_user:bhuvana1685@cluster0.bbj8cfx.mongodb.net/
  debtre relief?retryWrites=true&w=majority&appName=Cluster0
4
5  JWT_SECRET=DebtReliefSecret123
6  GEMINI_API_KEY=AQ.Ab8RN6J5fyoiAiYV-U3d1PaMzG27LVliIkCmT_QyXn_mvINmJQ
```

Step 7 - Run the Backend Server

```
</> Bash  
  
npm run server
```

Step 8 - Run the React Application

```
</> Bash  
  
npm start
```

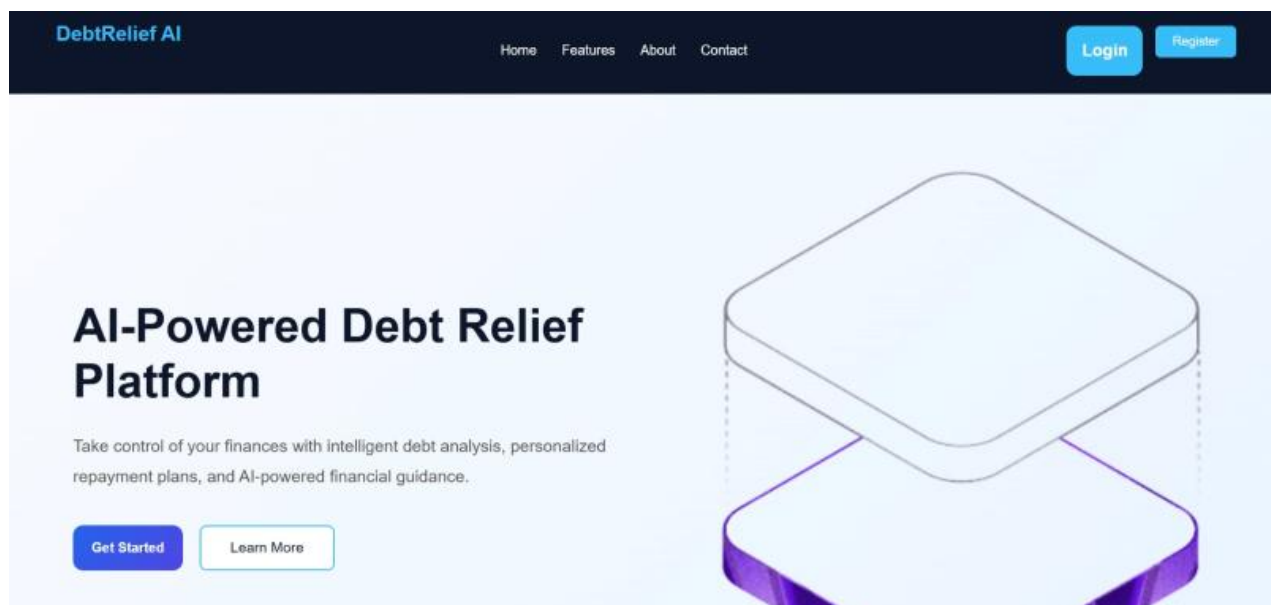
Open the application in your browser.

```
http://localhost:3000
```

Step 9 - Verify the Application

Verify that the following modules are functioning correctly:

- User Registration
- User Login
- Dashboard
- Add Loan
- Loan Management
- AI Analysis
- Analytics Dashboard
- Debt Recommendation Generation



Milestone 2: Core Functionalities Development

Milestone 2 focuses on implementing the core functionalities of the AI-Powered Debt Relief & Financial Recovery Platform. In this phase, the backend services, user authentication, loan management system, AI-powered financial analysis, and recommendation modules are developed. These functionalities enable users to securely manage their financial information and receive personalized debt relief recommendations generated by Google Gemini AI.

Activity 2.1: Develop Core Financial Management Features

Implement the core functionalities that allow users to manage their financial information and receive AI-powered assistance.

User Authentication:

- Develop secure user registration and login functionality.
- Encrypt user passwords using bcrypt.js before storing them in the database.
- Implement JWT-based authentication to protect user sessions and secure API access.

Loan Management

- Allow users to add multiple loan records.
- Store details such as loan type, outstanding amount, EMI, interest rate, due date, and lender information.
- Provide options to view, update, and delete loan records.

Financial Analytics

- Calculate the user's debt-to-income ratio, monthly EMI burden, outstanding balance, and repayment progress.
- Display financial insights using interactive charts and graphs.

AI Recommendation Generator

- Generate personalized debt settlement strategies based on the user's financial profile.
- Provide budgeting suggestions and repayment plans.
- Generate professional loan negotiation letters using Google Gemini AI.

Activity 2.2: Implement the Backend using Node.js and Express.js

Define REST API Routes

Develop RESTful API endpoints for:

- User Registration
- User Login
- Loan Management
- AI Financial Analysis
- Analytics Dashboard
- Negotiation Letter Generation

Process User Financial Data

- Receive financial information submitted from the React frontend.
- Validate all required fields before processing.
- Ensure loan amount, monthly income, monthly expenses, EMI, and interest rate contain valid values.
- Reject incomplete or invalid requests with appropriate error messages.

Database Integration

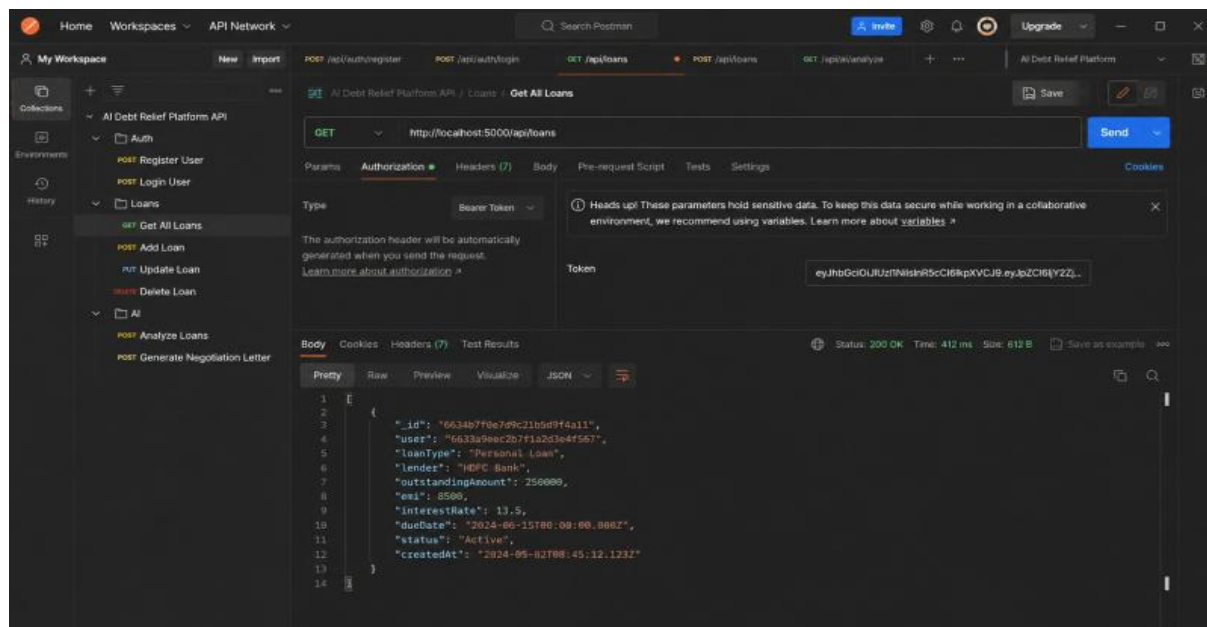
- Store user profiles in MongoDB Atlas.
- Store loan details securely using Mongoose.
- Retrieve financial information for dashboard analytics.
- Maintain AI-generated recommendations and negotiation letter history.

Integrate Google Gemini AI

- Send validated financial information to Google Gemini AI.
- Analyze the user's financial condition.
- Generate personalized:
 - Debt Settlement Recommendations
 - Repayment Plans
 - Budgeting Suggestions
 - Financial Health Insights
 - Loan Negotiation Letters
- Return AI-generated responses to the frontend in a structured JSON format.

Implement Error Handling

- Handle invalid user inputs gracefully.
- Manage database connection failures.
- Handle AI API request failures.
- Display meaningful error messages to users without exposing sensitive information.



Milestone 3: App.py Development

Milestone 3 focuses on developing the backend logic of the AI-Powered Debt Relief & Financial Recovery Platform. In this milestone, the core server-side functionalities are implemented using Node.js and Express.js. The backend manages user authentication, loan management, AI integration with Google Gemini, financial data processing, and communication with the MongoDB Atlas database. Proper API routing, request validation, and error handling are also implemented to ensure secure and reliable application performance.

Activity 3.1: Develop the Main Backend Application

Configure the Express Server: Develop the main server application (server.js) to initialize the Express application, configure middleware, establish database connectivity, and register all API routes.

The server performs the following tasks:

- Initializes the Express application.
- Connects to MongoDB Atlas.
- Configures middleware for JSON parsing and CORS.
- Loads environment variables.
- Registers authentication, loan management, analytics, and AI routes.
- Starts the backend server.

Define REST API Routes

Create RESTful API endpoints for different application modules.

Authentication Routes

Develop routes for:

- User Registration
- User Login
- JWT Authentication
- Protected User Access

Loan Management Routes

Develop routes for:

- Add Loan
- View Loans
- Update Loan
- Delete Loan

These routes allow authenticated users to securely manage their financial information.

AI Analysis Routes

Develop routes for:

- Debt Analysis
- AI Recommendation Generation
- Financial Health Assessment
- Negotiation Letter Generation

These APIs receive user financial data, communicate with Google Gemini AI, and return personalized recommendations.

Process User Requests

The backend receives user requests from the React frontend and performs the following operations:

- Validate all incoming request data.
- Authenticate users using JWT.
- Process financial information.
- Retrieve user loan details from MongoDB Atlas.
- Send financial data to Google Gemini AI.
- Return AI-generated recommendations in JSON format.

Connect with MongoDB Atlas

Using Mongoose, the backend stores and retrieves:

- User Profiles
- Loan Information
- Financial Records
- AI Recommendations
- Negotiation Letter History

Database operations include Create, Read, Update, and Delete (CRUD) functionalities.

Integrate Google Gemini AI

The backend communicates with Google Gemini AI to analyze users' financial information.

The AI generates:

- Personalized Debt Settlement Recommendations
- Repayment Strategies
- Budget Planning Suggestions
- Financial Health Insights
- Loan Negotiation Letters

The AI response is processed and returned to the frontend in a structured format.

Implement Request Validation

Before processing any request, the backend validates:

- User authentication
- Required financial fields
- Loan amount
- Monthly income
- Monthly expenses
- EMI amount
- Interest rate

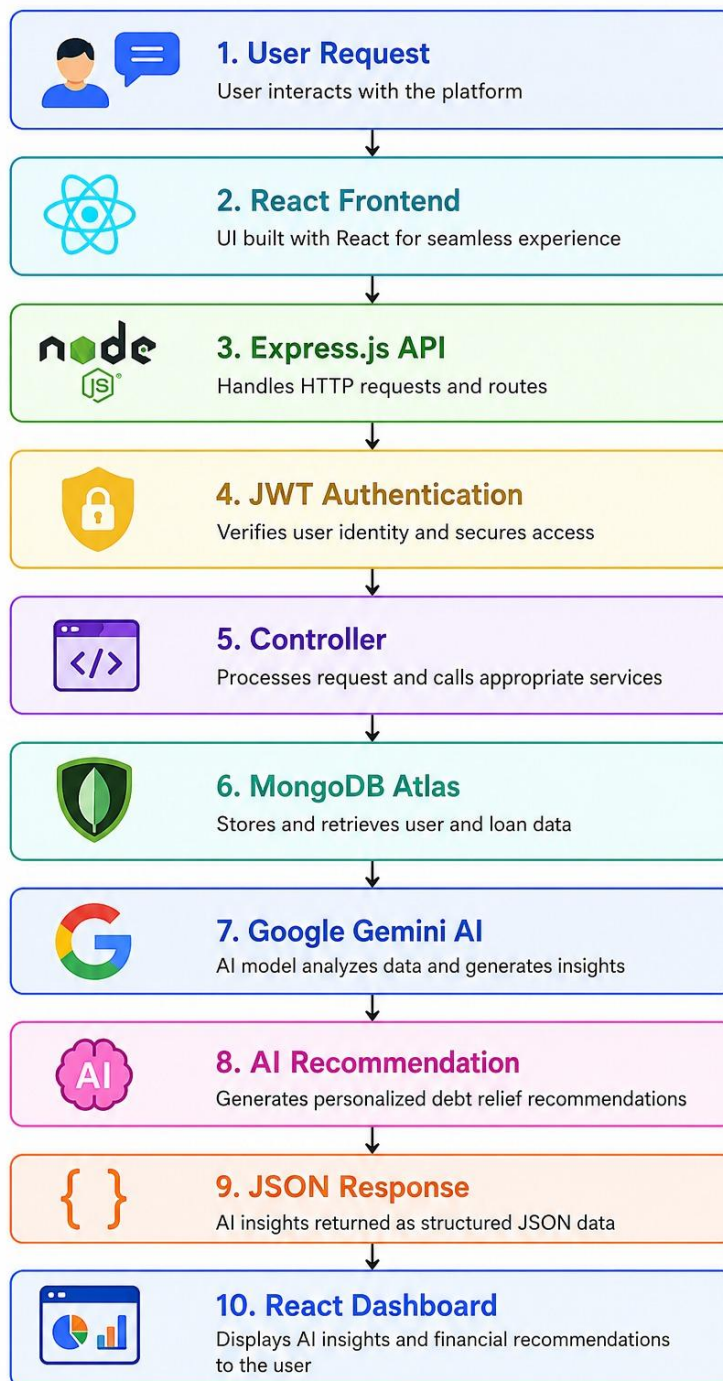
Invalid or incomplete requests return meaningful error messages.

Implement Error Handling

To improve application reliability, the backend handles:

- Invalid user inputs
- Authentication failures
- Database connection errors
- AI API failures
- Internal server exceptions

Appropriate HTTP status codes and error messages are returned to the frontend



Milestone 4: Frontend Development

In this milestone, the focus is on developing a responsive and user-friendly interface for the AI Debt Relief Platform. The frontend is built using React.js with modern UI components to provide users with an intuitive experience for loan management, AI-powered debt analysis, financial tracking, and personalized recommendations.

Activity 4.1: Designing and Developing the User Interface

Set Up the React Application

- Create the React project using Vite or Create React App.
- Organize the project into reusable components such as Navbar, Hero, Dashboard, Login, Register, Add Loan, My Loans, AI Analysis, and Analytics.
- Configure React Router for smooth navigation between pages.

Design a Responsive User Interface

- Develop a modern and responsive layout using CSS.
- Create a colorful dashboard with cards, charts, and interactive buttons.
- Ensure compatibility across desktop, tablet, and mobile devices.
- Add animations and hover effects for a better user experience.

Develop Authentication Pages

- Create Login and Registration forms.
- Validate user inputs before sending requests to the backend.
- Display success and error messages dynamically.

Create Dashboard Components

Develop separate UI sections for:

Loan Summary Cards

- Display Total Debt
- Monthly EMI
- Remaining Balance
- Debt Stress Level

Loan Management

- Add new loans
- View loan details
- Delete loans
- Update repayment information

AI Recommendation Panel

- Display AI-generated settlement suggestions.
- Show financial health score.
- Present personalized repayment strategies.

Analytics Dashboard

- Bar Charts
- Pie Charts
- Debt Distribution Graphs
- Monthly EMI Analysis

Activity 4.2: Creating Dynamic User Interaction

Connect Frontend with Backend APIs

- Use Axios to communicate with the Express.js backend.
- Fetch loan information dynamically.
- Display AI recommendations received from the backend.

Implement Protected Routes

- Store JWT token after successful login.
- Restrict dashboard access to authenticated users.
- Redirect unauthenticated users to the Login page.

Render Loan Data Dynamically

- Retrieve all loans from MongoDB Atlas.
- Display loan details using reusable React components.
- Refresh data automatically after adding or deleting loans.

Display AI Recommendations

- Send financial data to the backend.
- Receive AI-generated debt analysis.
- Display:
 - Recommended repayment strategy
 - Settlement percentage
 - Financial improvement tips
 - Monthly savings suggestions

Visualize Financial Analytics

- Use Chart.js to display:
 - Debt Breakdown
 - EMI Distribution

- Loan Progress
- Financial Health Score

Improve User Experience

- Show loading animations during API requests.
- Display toast notifications for successful operations.
- Handle errors gracefully with informative messages.
- Ensure fast and responsive page navigation.

Expected Frontend Pages

- Home Page
- Login Page
- Register Page
- Dashboard
- Add Loan Page
- My Loans Page
- AI Analysis Page
- Analytics Page
- User Profile Page (Optional)

Milestone 5: Deployment

In this milestone, the AI Debt Relief Platform is prepared for deployment by configuring the backend server, frontend application, cloud database, and AI services. The application is thoroughly tested in both local and cloud environments to ensure secure communication, reliable performance, and accessibility. After successful testing, the backend is deployed on Render, the frontend is deployed on Vercel, and MongoDB Atlas serves as the cloud database. This deployment enables users to securely access AI-powered debt analysis and financial recommendations from anywhere.

Activity 5.1: Preparing the Application for Deployment

Before deploying the application, the project environment was configured by installing all required dependencies and setting up the backend server. Environment variables were securely stored to protect sensitive information such as database credentials, JWT secret keys, and Google Gemini API keys.

Configure the Backend Environment

- Install all required Node.js dependencies using npm.
- Configure the Express.js server with middleware such as Express, CORS, dotenv, JWT, and Mongoose.
- Verify that all REST API endpoints are functioning correctly.

- Ensure controllers, routes, and middleware are properly connected.

Configure Environment Variables

Create a .env file and store all sensitive credentials securely.

Configure MongoDB Atlas

- Create a MongoDB Atlas cluster.
- Configure database users and passwords.
- Enable network access for deployment.
- Verify CRUD operations for Users and Loans collections.

Configure Google Gemini AI

- Generate a Google Gemini API key.
- Connect the backend server with Gemini AI.
- Test AI-generated debt analysis and settlement recommendations.
- Validate AI responses before returning them to the frontend.

Activity 5.2: Local Testing and Verification

Before cloud deployment, the application was tested locally to ensure all modules function correctly without errors.

Run the Backend Server:

```
npm run server
```

Run the React Frontend:

```
npm start
```

Verify Application Modules

- The following modules were tested successfully:
- User Registration
- User Login
- JWT Authentication
- Add Loan
- View Loans
- Delete Loan
- AI Debt Analysis

- Debt Settlement Recommendation
- Financial Health Dashboard
- Loan Analytics
- Logout

Validate API Communication

The following checks were performed:

- React frontend successfully communicates with the Express.js backend.
- JWT authentication secures protected routes.
- Loan information is stored and retrieved from MongoDB Atlas.
- Google Gemini AI generates personalized financial recommendations.
- Dashboard charts update dynamically based on stored loan data.
- All API responses are returned in JSON format without errors.

Activity 5.3: Cloud Deployment

After successful local testing, the application was deployed to cloud platforms to make it publicly accessible.

Deploy Backend using Render

The backend server was deployed on Render, which hosts the Express.js API and manages all server-side operations. Environment variables were configured, and the deployment status was monitored to ensure successful execution.

Deployment steps:

- Push backend code to GitHub.
- Connect the GitHub repository with Render.
- Configure environment variables.
- Deploy the Express.js backend.
- Verify API endpoints after deployment.

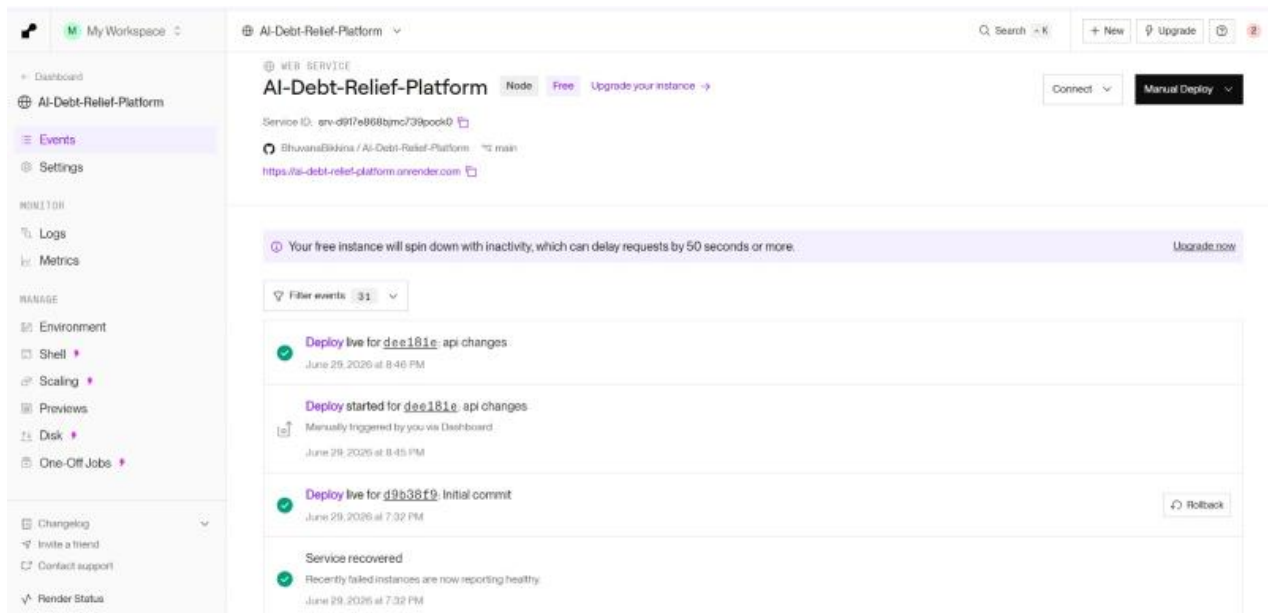


Figure 5.2: Express.js Backend Successfully Deployed on Render

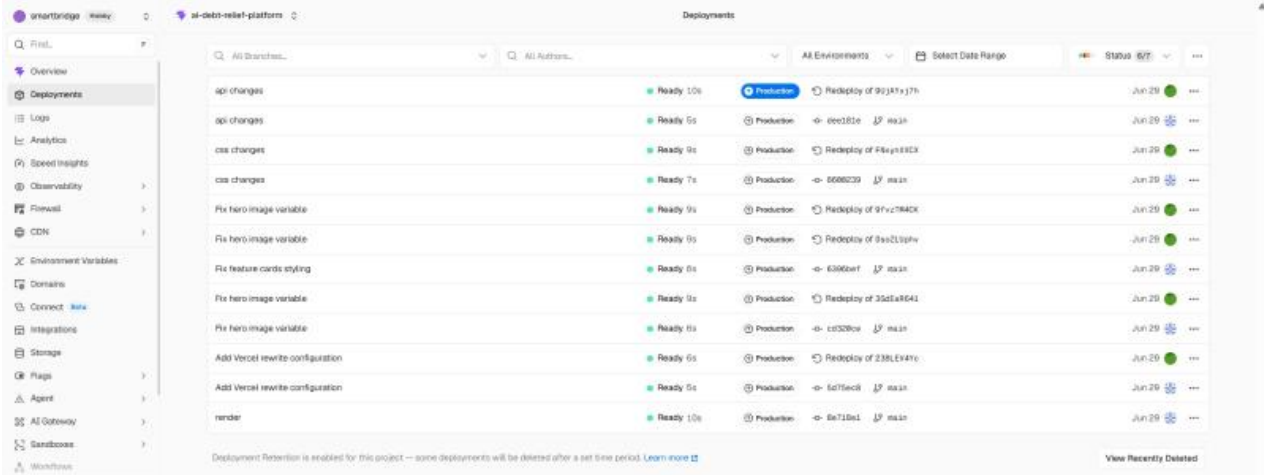
Deploy Frontend using Vercel

The React frontend was deployed using Vercel. After connecting the GitHub repository, the production build was automatically generated and deployed. The frontend was configured to communicate with the deployed backend API.

Deployment steps:

- Push React project to GitHub.
- Connect the repository with Vercel.
- Configure the production backend URL.
- Deploy the frontend.
- Verify all pages load correctly.

Figure 5.3: React Frontend Successfully Deployed on Vercel



Branch	Status	Environment	Deployment	Author	Time	Actions
api changes	Ready 10s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
api changes	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
css changes	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
css changes	Ready 7s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Fix hero image variable	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Fix hero image variable	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Fix feature cards styling	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Fix hero image variable	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Fix hero image variable	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Add Vercel rewrite configuration	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
Add Vercel rewrite configuration	Ready 5s	Production	Redeploy of 92387xj7h		Jun 29	Rollback
render	Ready 10s	Production	Redeploy of 92387xj7h		Jun 29	Rollback

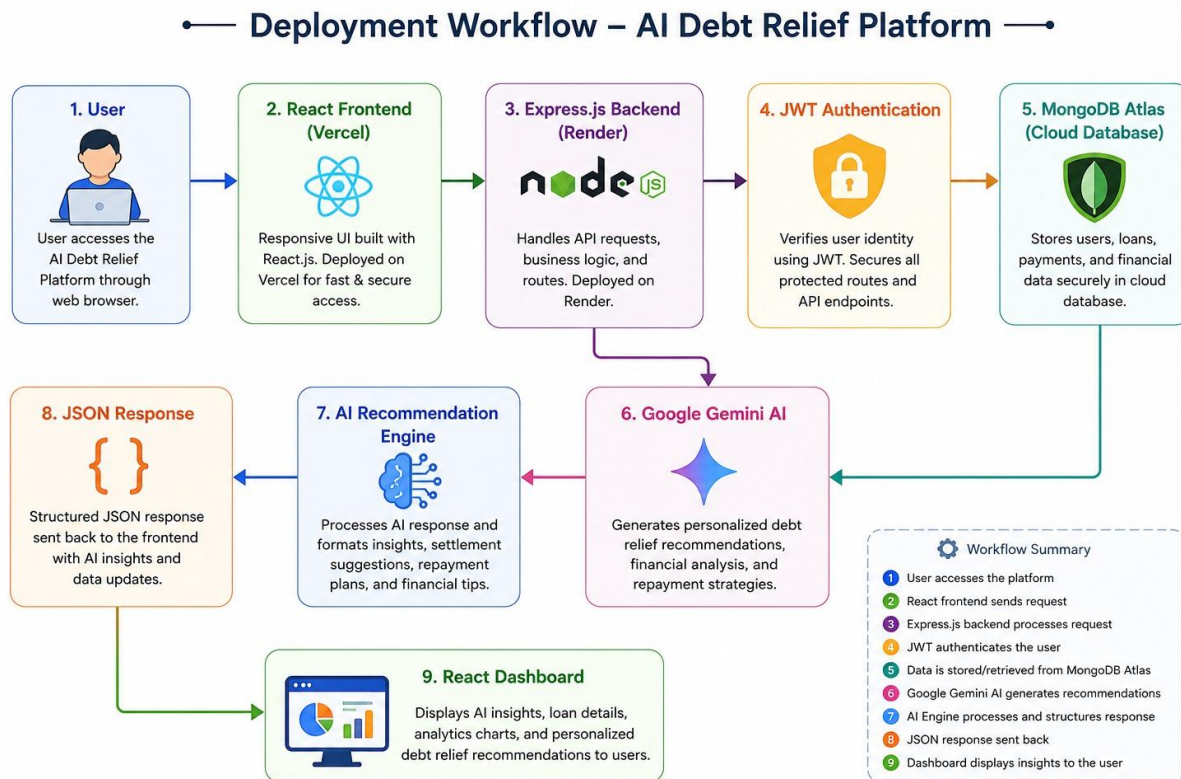
Final System Verification

After deployment, complete end-to-end testing was performed to verify the entire application.

The following functionalities were verified successfully:

- User Registration and Login
- Secure JWT Authentication
- Loan Management
- MongoDB Atlas Database Connectivity
- Google Gemini AI Integration
- AI Debt Recommendation Generation
- Dashboard Analytics
- Responsive User Interface
- Cross-Browser Compatibility
- API Performance and Response Time

Deployment Workflow:



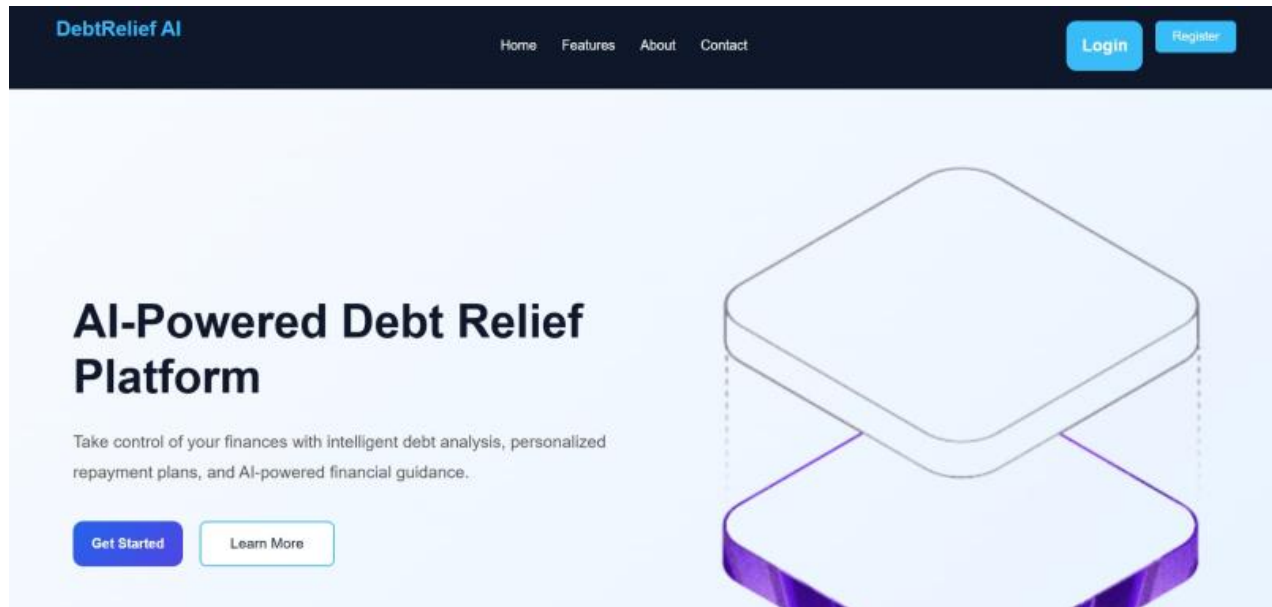
Exploring the Website's Web Pages:

1. Home Page

Description:

The Home Page serves as the landing page of the AI Debt Relief Platform. It provides an attractive introduction to the application with a modern and responsive design. The page includes a navigation bar, hero section, feature highlights, and call-to-action buttons that guide users to register or log in. It also presents the platform's objective of helping users analyze their debt and receive AI-powered financial recovery recommendations.

Figure 6.1: AI Debt Relief Platform Home Page

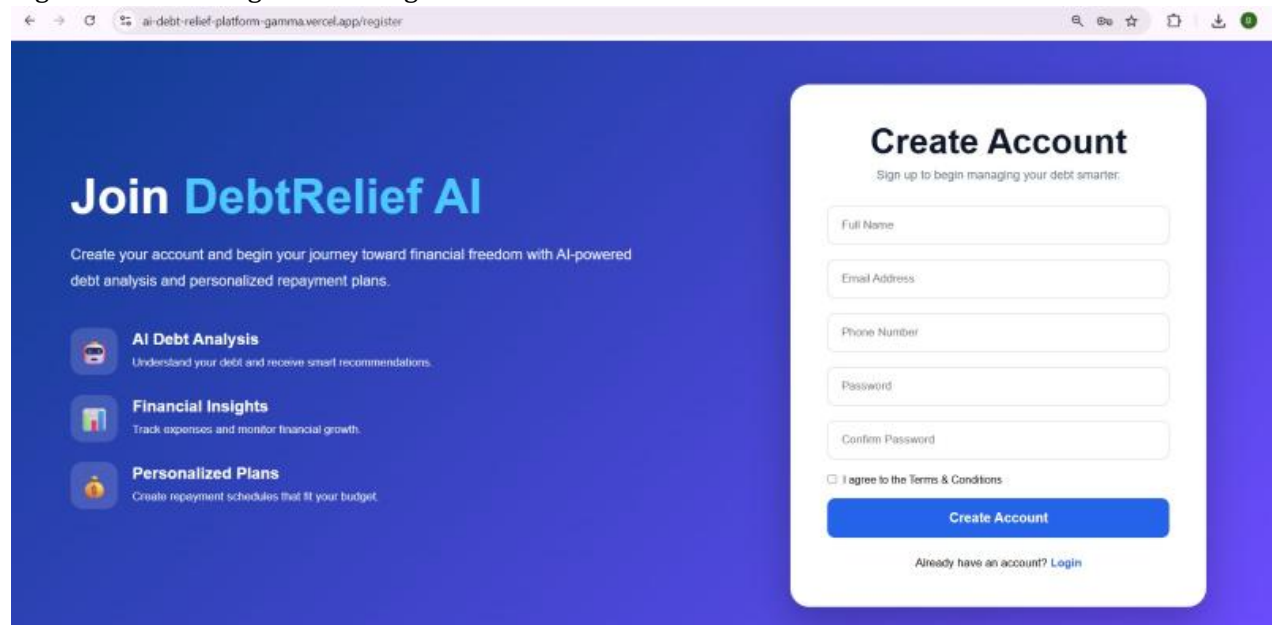


2. User Registration Page

Description:

The Registration page enables new users to create an account by entering personal details such as name, email address, and password. Input validation ensures that all required information is provided before securely storing user data in the MongoDB Atlas database. After successful registration, users are redirected to the login page.

Figure 6.2: User Registration Page



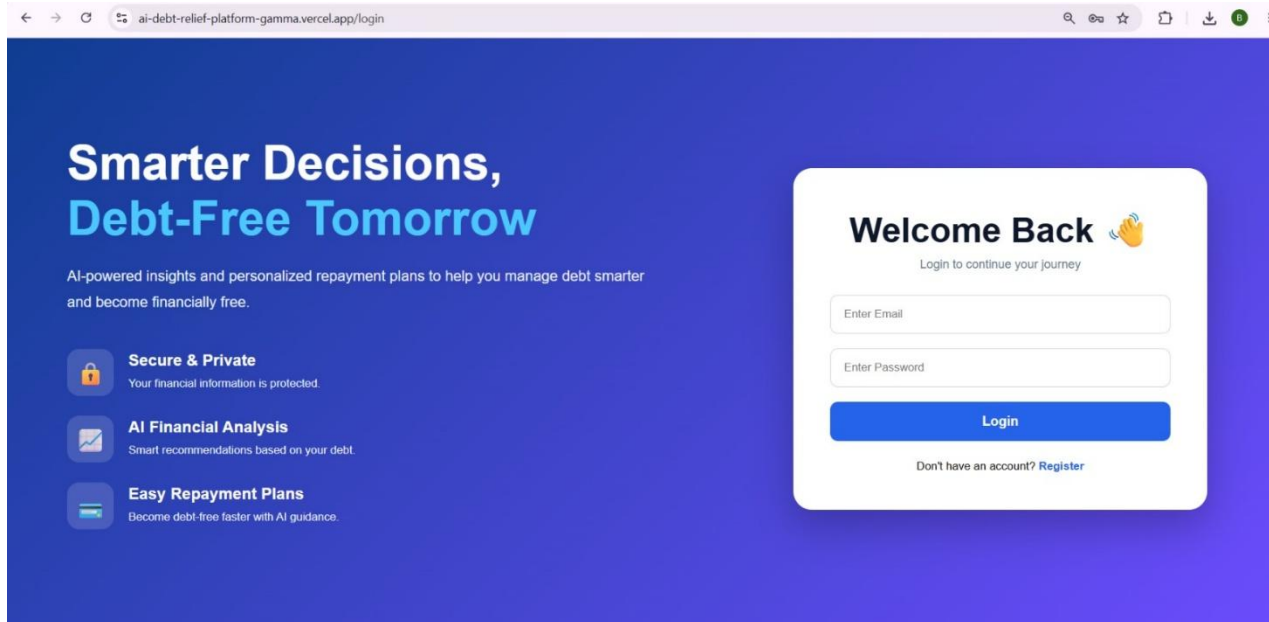
The image is a screenshot of the 'Create Account' page in a web browser. The browser's address bar shows 'ai-debt-relief-platform-gamma.vercel.app/register'. The page has a dark blue background. On the left, the heading 'Join DebtRelief AI' is followed by the text: 'Create your account and begin your journey toward financial freedom with AI-powered debt analysis and personalized repayment plans.' Below this are three feature cards: 'AI Debt Analysis' (Understand your debt and receive smart recommendations), 'Financial Insights' (Track expenses and monitor financial growth), and 'Personalized Plans' (Create repayment schedules that fit your budget). On the right, there is a white card titled 'Create Account' with the subtext 'Sign up to begin managing your debt smarter.' It contains input fields for 'Full Name', 'Email Address', 'Phone Number', 'Password', and 'Confirm Password'. Below these fields is a checkbox for 'I agree to the Terms & Conditions' and a blue 'Create Account' button. At the bottom of the card, it says 'Already have an account? Login'.

3. Login Page

Description:

The Login page allows registered users to securely access the platform using their email and password. JWT authentication is used to verify user credentials and protect restricted pages. After successful authentication, users are redirected to the dashboard.

Figure 6.3: User Login Page

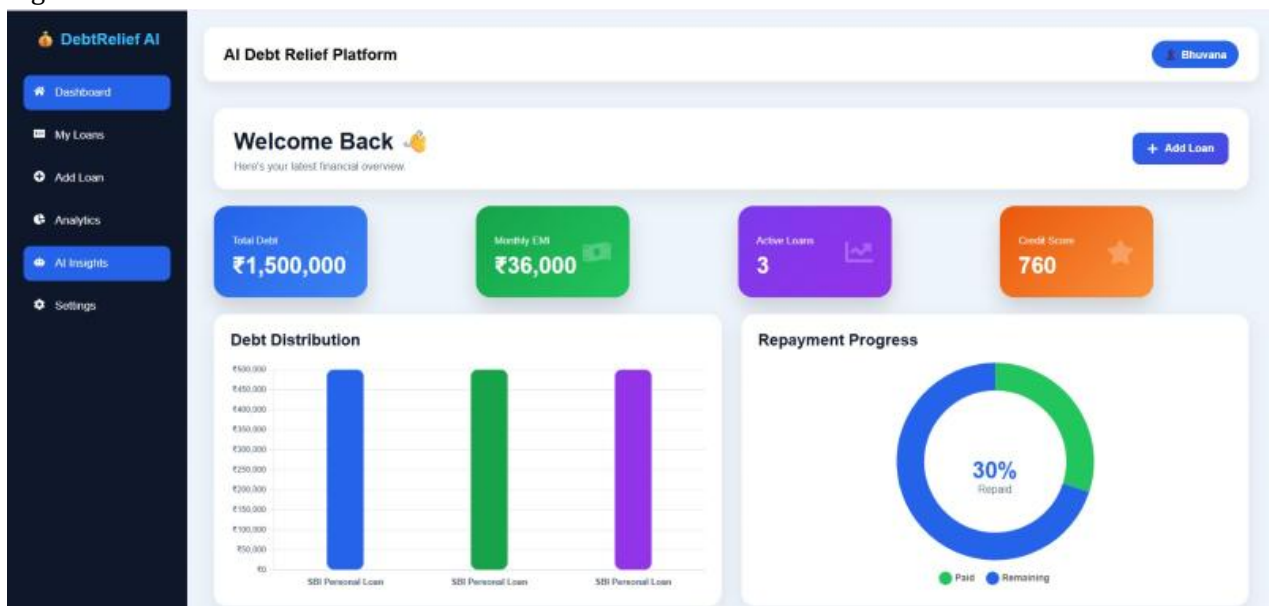


4. Dashboard

Description:

The Dashboard provides an overview of the user's financial information. It displays important statistics such as total outstanding debt, monthly EMI, debt status, and repayment progress. Users can quickly access loan management, AI analysis, and financial analytics from this centralized interface.

Figure 6.4: Financial Dashboard

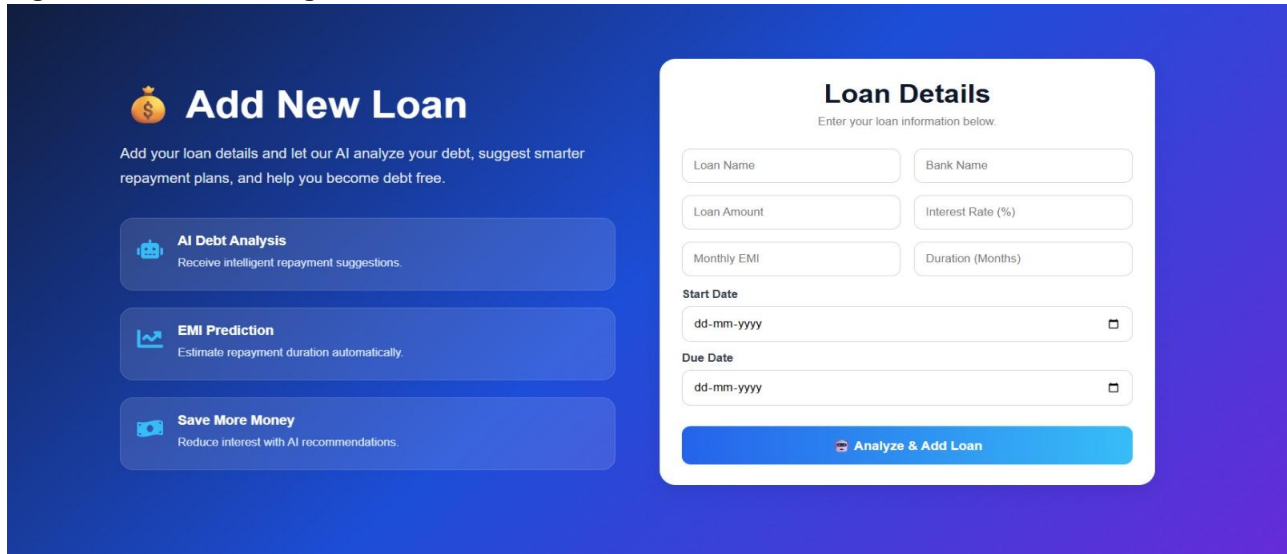


5. Add Loan Page

Description:

The Add Loan page allows users to enter their loan details, including loan amount, interest rate, monthly EMI, loan tenure, and lender information. The submitted data is securely stored in MongoDB Atlas and becomes available for AI analysis and dashboard visualization.

Figure 6.5: Add Loan Page



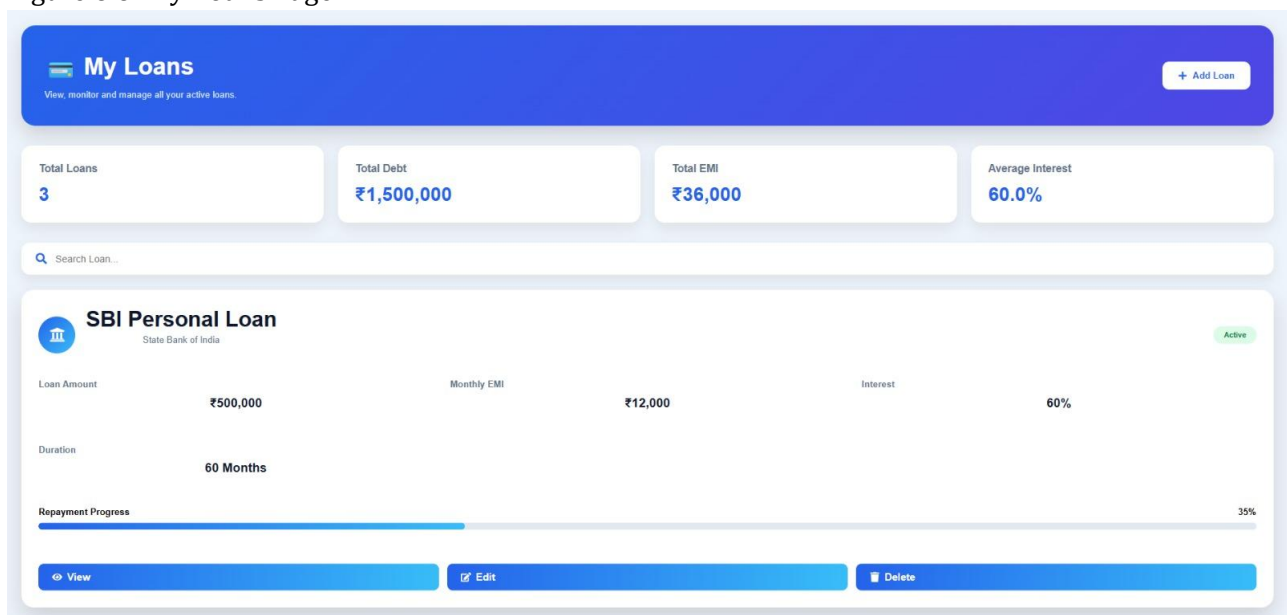
The screenshot shows the 'Add New Loan' page. On the left, there's a blue sidebar with three cards: 'AI Debt Analysis' (Receive intelligent repayment suggestions), 'EMI Prediction' (Estimate repayment duration automatically), and 'Save More Money' (Reduce interest with AI recommendations). The main area is white and titled 'Loan Details' with the subtitle 'Enter your loan information below.' It contains several input fields: 'Loan Name', 'Bank Name', 'Loan Amount', 'Interest Rate (%)', 'Monthly EMI', 'Duration (Months)', 'Start Date' (dd-mm-yyyy), and 'Due Date' (dd-mm-yyyy). At the bottom right is a blue button labeled 'Analyze & Add Loan'.

6. My Loans Page

Description:

The My Loans page displays all loans associated with the authenticated user. Users can review loan details, monitor repayment status, and delete loans if required. The page dynamically retrieves loan information from the backend using REST APIs.

Figure 6.6: My Loans Page



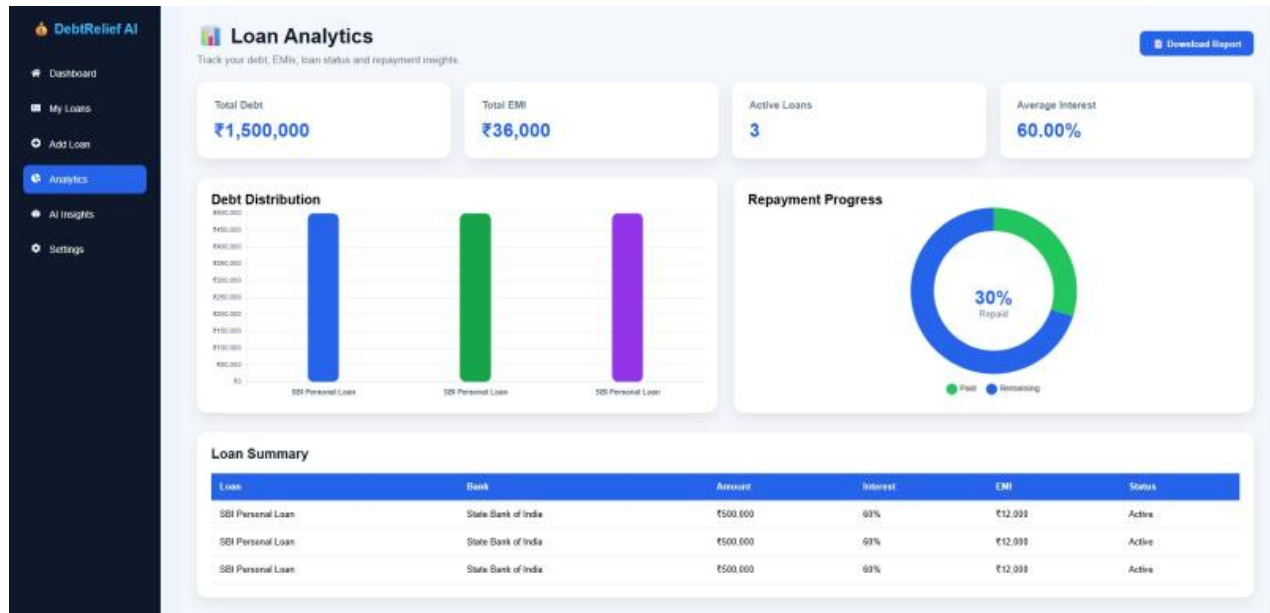
The screenshot shows the 'My Loans' page. At the top is a blue header with the title 'My Loans' and a subtitle 'View, monitor and manage all your active loans.' There's a '+ Add Loan' button on the right. Below the header are four summary cards: 'Total Loans' (3), 'Total Debt' (₹1,500,000), 'Total EMI' (₹36,000), and 'Average Interest' (60.0%). A search bar is below these cards. The main section displays a loan card for 'SBI Personal Loan' (State Bank of India), which is 'Active'. The card shows 'Loan Amount' (₹500,000), 'Monthly EMI' (₹12,000), 'Interest' (60%), and 'Duration' (60 Months). A 'Repayment Progress' bar shows 35% completion. At the bottom of the card are three buttons: 'View', 'Edit', and 'Delete'.

7. Analytics Page

Description:

The Analytics page presents graphical insights into the user's financial data using interactive charts. Users can visualize debt distribution, monthly EMI, repayment progress, and overall financial health. These visualizations help users understand their financial position and make informed repayment decisions.

Figure 6.7: Financial Analytics Dashboard

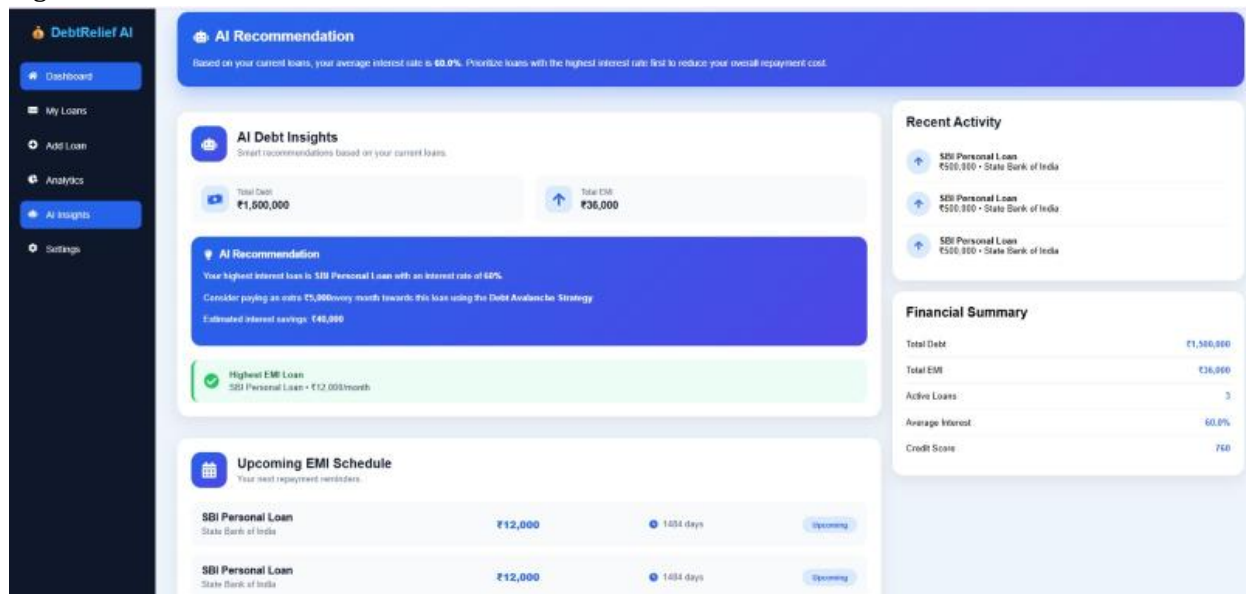


8. AI Recommendation Results

Description:

After analyzing the financial information, the platform displays AI-generated recommendations in an easy-to-understand format. The recommendations include debt settlement options, monthly savings suggestions, financial improvement tips, and repayment priorities, enabling users to make informed financial decisions.

Figure 6.8: AI-Powered Financial Recommendation



Conclusion:

The AI Debt Relief Platform is an intelligent financial assistance system designed to help users effectively manage their debts and improve their financial well-being through AI-powered insights. By integrating React.js for the frontend, Express.js for the backend, MongoDB Atlas for cloud-based data storage, JWT Authentication for secure user access, and Google Gemini AI for financial analysis, the platform provides users with personalized debt repayment strategies, financial health assessments, and settlement recommendations.

Throughout the project, various milestones were successfully completed, including system architecture design, backend development, AI integration, frontend implementation, testing, and cloud deployment. The application enables users to securely manage loan information, visualize financial analytics through interactive dashboards, and receive intelligent recommendations tailored to their financial situation.

The successful deployment of the application on cloud platforms ensures accessibility, scalability, and reliable performance while maintaining data security and efficient API communication. The platform demonstrates how Generative AI can simplify complex financial decision-making and support users in planning effective debt recovery strategies.

In the future, the AI Debt Relief Platform can be enhanced by integrating real-time credit score analysis, automated EMI reminders, multi-language support, secure online payment integration,

document upload for loan verification, predictive financial forecasting, and mobile application support. These enhancements will further improve user experience and make the platform a comprehensive AI-powered financial management solution.

Overall, the AI Debt Relief Platform showcases the practical application of modern web technologies and Artificial Intelligence in solving real-world financial challenges, providing users with a secure, reliable, and intelligent system for achieving long-term financial stability.